RAGNIFY Engineering & SaaS Architecture Standards (2026)

Security Classification: INTERN_PUBLIC | Authorized Roles: INTERN,EMPLOYEE,MANAGER,HR,ADMIN

RAGNIFY Engineering & SaaS Architecture Guide 2026 — Section 1: Production Technology Stack.
• Frontend Architecture: React 18, Vite 8, React Router v6, CSS Custom Properties Design System (Light & Dark SaaS themes), and Axios service layer with graceful offline fallbacks.
• Backend Architecture: Python FastAPI, SQLAlchemy ORM, PostgreSQL relational database (employees, tasks, onboarding, documents, working hours, tenures, notifications, and meetings).
• AI & RAG Pipeline: ChromaDB PersistentClient vector store ('ragnify_docs' collection), SentenceTransformer ('all-MiniLM-L6-v2' 384-dimensional dense embeddings), Groq Cloud LLM ('openai/gpt-oss-120b'), Whisper speech-to-text for meeting audio transcription, and ReportLab + python-pptx for automated PDF and PowerPoint generation.

RAGNIFY Engineering & SaaS Architecture Guide 2026 — Section 2: Sprint Delivery & Code Quality Rules.
• Automated Task Progress: Team and individual project progress percentages are calculated automatically from completed tasks (Completed Tasks / Total Assigned Tasks × 100). Managers assign, edit, or cancel tasks, while progress updates strictly when employees/interns complete deliverables.
• Pull Request & Deployment SLA: Every feature branch requires passing 'npm run build' with zero warnings/errors, unit test validation, and code review sign-off by the Engineering Manager within 24 hours.